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EDUCATION

Accelerated Master of Science in Biomedical Engineering | Expected May 2027

Bachelor of Science in Biomedical Engineering | May 2026

The University of Arizona, College of Engineering | Tucson, Arizona

GPA: 3.43

Relevant Coursework: BME 586: Biomaterial-Tissue Interactions, BME 488: Bioprinting of Functional Tissues, BME 461: Biological and Synthetic Materials, BME585: Nanotechnology for Biomedical Engineers, BME 516: Biomedical Imaging, BME 376: Biomedical Statistics

Research Programs and Honors: Undergraduate Biology Research Program (UBRP) Scholar; Place-based Health Informatics Research Experience (PHIRE) | Dean's List Spring/Fall 2024 | Wildcat Distinction Tuition Award

PROFESSIONAL EXPERIENCE

Research Assistant | May 2024 – Present

Dr. Hutchinson's Multi-scale Brain Imaging Lab, University of Arizona | Tucson, Arizona

Ferret Cradle Design Project – Research Equipment Development

- Engineered MRI-compatible medical device prototype by translating user needs from veterinary staff and MRI technicians into functional specifications in **SolidWorks**, and validated performance on 7T Bruker scanner.
- Performed iterative prototyping and validation on a 7T MRI system, ensuring compatibility, reliability, and repeatable performance.
- Designed a modular rail-and-hook attachment system that improved upon commercial solutions, increasing usability and adaptability for multi-modal imaging.
- Collaborated with cross-functional teams (engineering, veterinary, imaging staff) to refine device design and meet operational constraints.

Ferret Behavior Project – Automation & Data Analysis

- Developed automated machine learning pipeline using **DeepLabCut** to classify animal behavior, reducing manual video annotation time and enabling high-throughput analysis of videos across 23 animal subjects.
- Created Python tools to extract kinematic metrics and analyze behavioral datasets, improving efficiency and enabling quantitative research insights.
- Designed custom **3D models** in **SolidWorks** for experimental apparatus, ensuring compliance with animal safety protocols and supporting successful execution of research trials.
- Conducted comprehensive literature reviews to identify research gaps and inform future experimental design.

RELEVANT LABORATORY AND ENGINEERING PROJECTS

Mechanical Engineering Lead-MACRO: Multidigit Assessment of Choice Response Output | Fall 2025 - Spring 2026

ENGR498A/B - Interdisciplinary Capstone | University of Arizona

- Supported design and development of an electromechanical hand assessment device integrating sensors (FSRs) and actuators (LRAs), guaranteeing a standardized and reliable procedure.
- Led **mechanical design** of an ergonomic, digit-size adjustable hand block for multidigit choice response assessment system, accommodating 95% of the population.

Medical Device Design – Heat Illness Detection Wearable | Spring 2025

BME 310: Medical Device Design | University of Arizona

- Integrated pulse-oximetry, accelerometer, and temperature sensors and used systematic root-cause analysis to diagnose communication errors, verify voltages, isolate faulty components, and complete circuit-level rework.

Bioprinting of Functional Tissues Laboratory | Fall 2025

BME 488, University of Arizona

- Cultured and passaged cells using sterile/aseptic technique during tissue-engineering and bioprinting laboratory exercises, following established protocols and clean-work practices.

RESEARCH TRAINING AND CERTIFICATIONS

CITI Program: Biomedical Research Investigators (expires Feb 2029) | Responsible Conduct of Research-Basic (expires May 2027) | Working with the IACUC-Investigators, Staff and Students | Working With Ferrets in Research Settings

University of Arizona: Science Communication-Undergraduate Biology Research | Research Readiness

TECHNICAL SKILLS

Laboratory and Research: Cell culture and passaging (course-based), experimental data collection, literature review, technical documentation, and troubleshooting

Data Analysis and Programming: Python, MATLAB, R, C++, Excel, DeepLabCut

Design and Prototyping: SolidWorks, 3D printing, soldering, Eagle PCB, hardware integration

Collaboration and Communication: Cross-functional research teams, scientific presentations, technical reports, fluent Spanish